Answer Key & Teacher Notes

Session 1 — Equations from Tables · Session 2 — Equations from Situations

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	About 10 minutes; use items 1–4 for a 5-minute check. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 · TEACHER NOTES

Equations from Tables

TALK PROMPTS

- How does a table help you spot the constant rate in a relationship?
- How can you tell from the table alone that a rule multiplies instead of adds?

PARTNER TASK

Choir Risers. A table shows x: 1, 2, 3, 4 risers and y: 7, 14, 21, 28 singers. One partner writes the equation; the other tests it on two rows and uses it to find the singers on 11 risers (77).

WATCH FOR

- Writing a rule that fits only the first row (y = x + 24 for x: 3, 5, 10 and y: 27, 45, 90).
- Switching the variables (x = 7y instead of y = 7x).
- Using the jump in y between rows as the rate when x does not go up by 1.

SESSION 2 · TEACHER NOTES

Equations from Situations

TALK PROMPTS

- How are the equation and the graph two views of the same relationship?
- What part of Zuri's equation would change if she raised her price to $24 for each car?

PARTNER TASK

Hall and Bounce House. The hall charges a $75 fee plus $30 an hour; the bounce house charges a $60 fee plus $25 an hour. Each partner writes one equation and finds the cost for 4 hours (hall $195, bounce house $160), then together compare the fees and the hourly rates.

WATCH FOR

- Adding the rate instead of multiplying (c = n + 2 for buttons that cost $2 each).
- Multiplying the fee by the variable (c = 75h + 30 instead of c = 30h + 75).
- Combining the fee and the rate into one rate (c = 105h).

ITEM ANALYSIS

suggested DOK; judge the reasoning required, not the question label

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.EE.C.9	1	2	1	1	2	3	2	3	2	1	1
S1 · Form B	6.EE.C.9	1	2	1	1	2	3	2	3	2	1	1
S2 · Form A	6.EE.C.9	1	2	1	2	1	3	2	2	2	1	1
S2 · Form B	6.EE.C.9	1	2	1	2	1	3	2	2	2	1	1

Items before the last three are Practice; then Find the Error (explain and correct) and two Test Practice items.

GUIDED NOTES

equation: A math sentence with an **equal** sign.

constant: A number that **never** changes.

rate: The amount for **each** one unit.

- If $y \div x$ is the **same** in every row, the equation is $y =$ (that number) $\times x$.
- If $y - x$ is the same in every row, the equation is $y = x$ **+** (that number).
- Check the equation with **two** rows.

Example: $6 \div 1 = \mathbf{6}$, $12 \div 2 = 6$, $30 \div 5 = 6$ The rate is **6** in every row. Answer: $y = \mathbf{6}x$. Check: $6 \times 5 = 30$.

YOU TRY

1. a. $y = 5x$ b. $y = x + 8$

2. $y = 8x$; $8 \times 12 = 96$

EXIT TICKET

1. a. $y = 4x$ b. $y = x + 5$

2. $y = 5x$; $5 \times 13 = 65$ cans

3. **B** (every row)

PRACTICE FORM A

1. a. $y = 5x$ b. $y = 12x$ c. $y = x + 3$ d. $y = 9x$

2. $y = 8x$; 7 passes cost $8 \times 7 = \$56$

3. $y = 5x$; $5 \times 12 = 60$ tiles

4. $y = 30x$; x = gallons of gas

5. $y = 7x$; $7 \times 14 = \$98$

6. For $x = 5$: $5 + 24 = 29$, not 45, so it only fits the first row. Each y is 9 times x: $y = 9x$.

7. $y = 25x$; x = number of packs, y = number of beads

8. y goes up 4 each time, and $x = 0$ would give 5: $y = 4x + 5$. When $x = 10$, $y = 4 \times 10 + 5 = 45$.

9. He multiplied the wrong variable. Each y is 7 times x: $7 \times 2 = 14$. The correct equation is $y = 7x$.

10. **B** ($y = 5x$)

11. **C** ($y = 3x + 1$)

Self-check word **FACTOR**

PRACTICE FORM B

1. a. $y = 6x$ b. $y = 11x$ c. $y = x + 7$ d. $y = 4x$

2. $y = 9x$; 8 passes cost $9 \times 8 = \$72$

3. $y = 7x$; $7 \times 11 = 77$ singers

4. $y = 24x$; x = cases of bottles

5. $y = 4x$; $4 \times 21 = \$84$

6. For $x = 4$: $4 + 14 = 18$, not 32, so it only fits the first row. Each y is 8 times x: $y = 8x$.

7. $y = 6x$; x = number of stacks, y = number of lockers

8. y goes up 3 each time, and $x = 0$ would give 4: $y = 3x + 4$. When $x = 10$, $y = 3 \times 10 + 4 = 34$.

9. He multiplied the wrong variable. Each y is 8 times x: $8 \times 3 = 24$. The correct equation is $y = 8x$.

10. **C** ($y = 12x$)

11. **A** ($y = 2x + 5$)

Self-check word **VALUES**

SESSION 2 · ANSWERS

GUIDED NOTES

unit rate: The amount for exactly **1** unit.

multiplicative rule: **Multiply** the input by the same number.

additive rule: **Add** the same number to the input.

per: Means "for **each** one."

- A rate ("each," "per") **multiplies** the variable.
- A one-time fee or a starting amount is **added** once.

Example: Define the letters: *d* = days, *e* = **dollars earned**. The rate is $9 for each **day**, so *e* = **9d**. Answer: Test *d* = 5: *e* = 9 × 5 = **45** dollars.

YOU TRY

1. a. p = 8b b. c = 20m + 10

2. p = 13n; 13 × 6 = $78

EXIT TICKET

1. a. c = 6t b. k = j + 4

2. c = 35n; 35 × 9 = $315

3. **B** (*c* = 10*h* + 25)

PRACTICE FORM A

1. a. c = 11h b. c = 14n c. m = b + 7 d. c = 5h + 20

2. a. independent: hours, h; dependent: cost, c b. c = 9h c. 9 × 6 = $54

3. p = 13n; 13 × 8 = $104

4. m = b + 7; 15 + 7 = 22 years old

5. r = 48b; 48 × 6 = 288 riders

6. A: y = 5x (multiply by 5). B: y = x + 4 (add 4). A multiplies x by a rate; B adds the same amount to x.

7. a. s = 45p b. s = 45p + 60 c. 45 × 4 + 60 = $240

8. c = 30h + 75; 30 × 5 + 75 = $225

9. Her rule adds 2 once. Test 5 buttons: 5 + 2 = 7, but 5 buttons cost 2 × 5 = $10. The price repeats for each button, so multiply: c = 2n.

10. **B** (*c* = 9*h*)

11. **C** (*c* = 5*h* + 20)

Self-check word **SYMBOL**

PRACTICE FORM B

1. a. c = 12h b. c = 15t c. l = s − 4 d. c = 8h + 30

2. a. independent: hours, h; dependent: cost, c b. c = 11h c. 11 × 7 = $77

3. e = 21n; 21 × 7 = $147

4. m = c + 5; 12 + 5 = 17 years old

5. r = 14v; 14 × 8 = 112 riders

6. A: y = 3x (multiply by 3). B: y = x + 2 (add 2). A multiplies x by a rate; B adds the same amount to x.

7. a. s = 35p b. s = 35p + 50 c. 35 × 6 + 50 = $260

8. c = 25h + 60; 25 × 4 + 60 = $160

9. Her rule adds 3 once. Test 4 stickers: 4 + 3 = 7, but 4 stickers cost 3 × 4 = $12. The price repeats for each sticker, so multiply: c = 3n.

10. **C** (*c* = 14*n*)

11. **A** (*c* = 6*h* + 15)

Self-check word **SOLVED**

HOMEWORK PACKET · ANSWERS

same answers for the core packet and Level 1

SHEET 1 · EQUATIONS FROM TABLES

1. a. y = 7x b. y = x + 9 c. y = 5x d. y = x − 3 e. y = x + 6 f. y = 9x g. y = 8x h. y = x − 5

2. y = 11x; 66, 99

3. Each y is 2 less than x: y = x − 2. When x = 20, y = 18.

4. y = 4x; 10 tickets cost $40

5. It only works for the first row: 3 × 4 = 12, not 8. Each y is 4 more than x, so y = x + 4.

SHEET 2 · EQUATIONS FROM SITUATIONS

1. a. y = 12x b. y = x + 3 c. y = 15x + 20

2. y = 5x + 4; 4 games: 5 × 4 + 4 = $24

3. y = 2x + 3; 7 miles: 2 × 7 + 3 = $17

4. $30, $38, $46, $70; y = 8x + 30

5. She added the fee to the rate, so every hour costs $35. The fee is paid once: y = 10x + 25.

6. **B** (*c* = 25*h* + 40)

7. **C** (Start with 12 gal, add 3 gal per min)

October 2026 conceptual additions — teacher key

Session 1 · Equations from Tables

A: y=5x. B: y=x+5. These rules fit the stated multiplicative/additive model families. Finite rows alone do not uniquely determine every conceivable rule.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: A: y=5x. B: y=x+5. These rules fit the stated multiplicative/additive model families. Finite rows alone do not uniquely determine every conceivable rule.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.

Session 2 · Equations from Situations

e=8v+5; v is visits, e dollars earned. The 8v term repeats per visit; 5 is added once. At v=3, e=29.

Extension: Accept a mathematically valid contrasting example and an explanation that uses the conditions in this session. Teacher anchor: e=8v+5; v is visits, e dollars earned. The 8v term repeats per visit; 5 is added once. At v=3, e=29.

Scoring: 1 point for a valid representation/setup, 1 for accurate mathematics and units, 1 for an explanation connecting the method to the task. Accept equivalent reasoning and suitable oral or diagram responses. Do not infer mastery from confidence alone.